4/27 CptS 350.

Final Exam Review I.

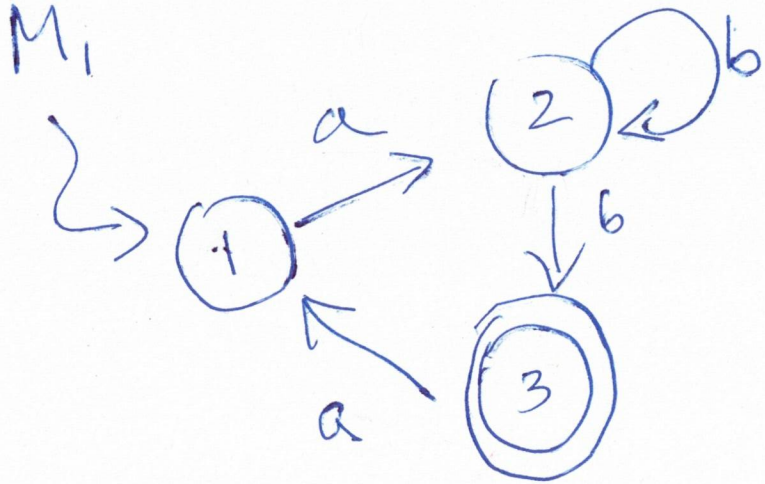
1. Do teaching Eval.
2. Prepare 2-page Single-sided cheat sheets.
   ↑ spend a lot of time.
   you can work in a group.
3. Watch this video & do all the problems.
   Re-do all standard hw problems.
4. Final covers all materials after midterm

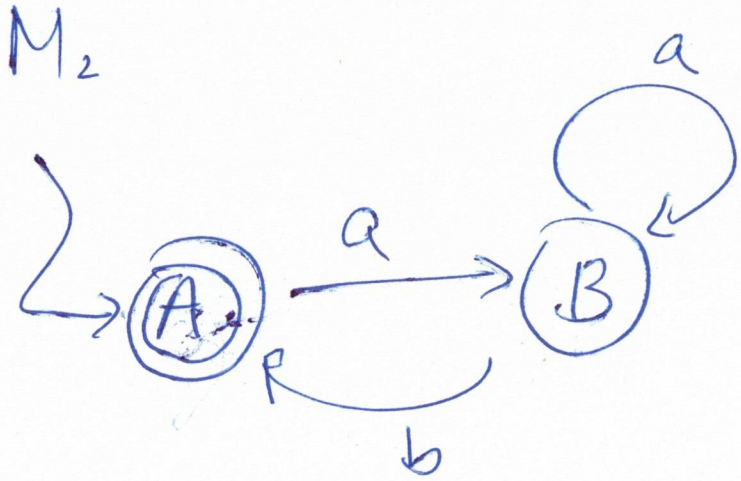
(review of Cartesian Product construction in 3.17).



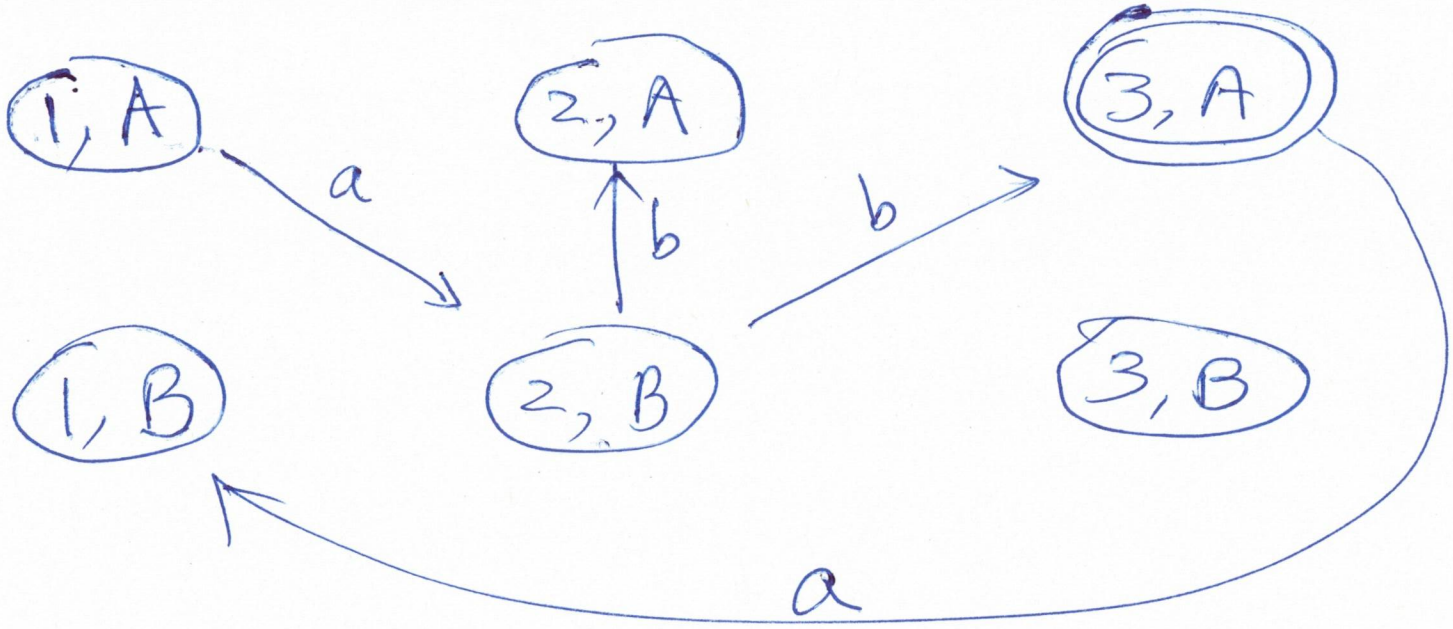
FA M_1 × FA M_2 = FA M

↑ Cartesian Product.





$M = M_1 \times M_2$



LCS with constraints:

Set-up: Given two strings α, β.
and a pattern as a regExp such as $(a+bb)^*cd$.

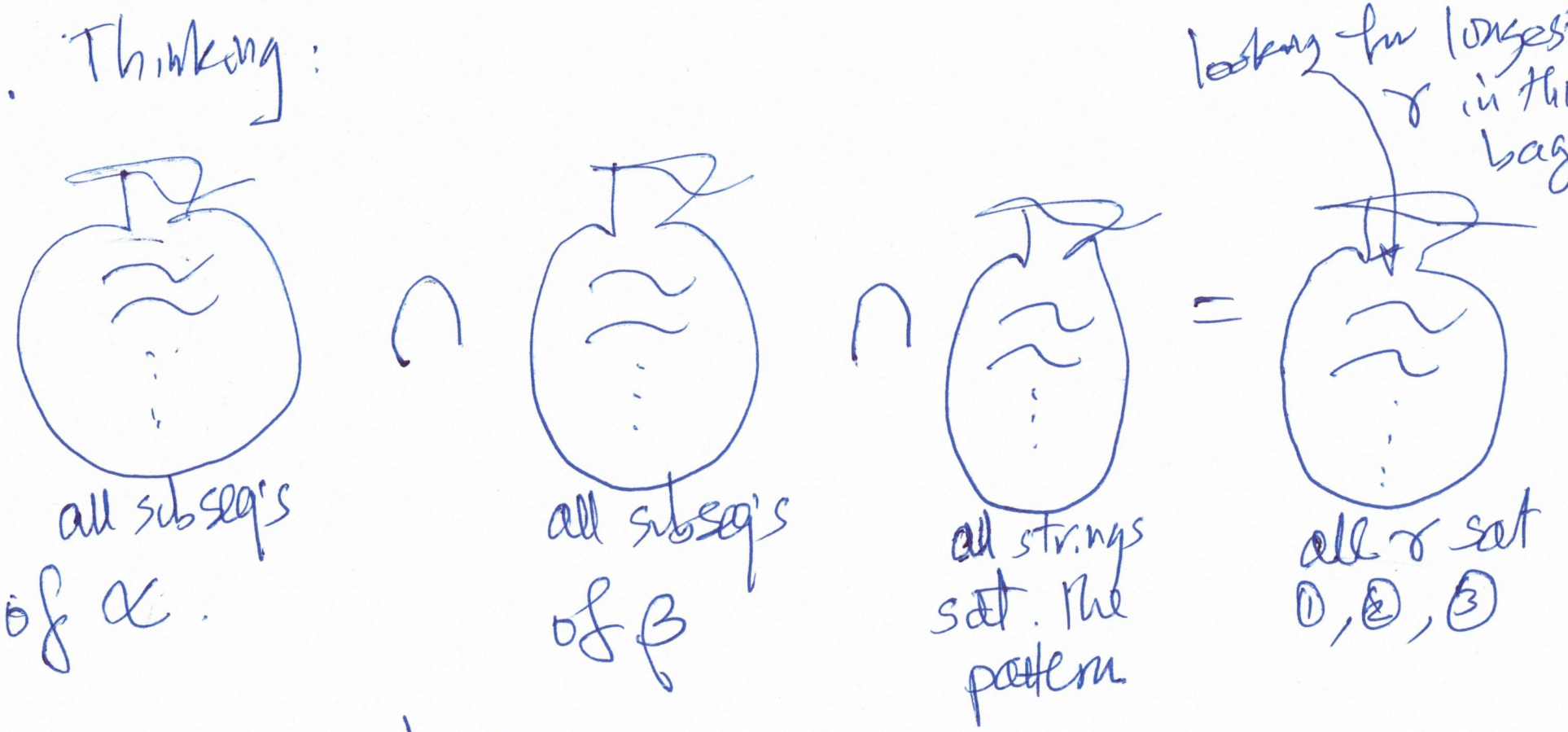
want: Find a longest γ s.t.

① γ is a subseq of α

② γ is a subseq of β

③ γ sat the pattern.

1. Thinking:



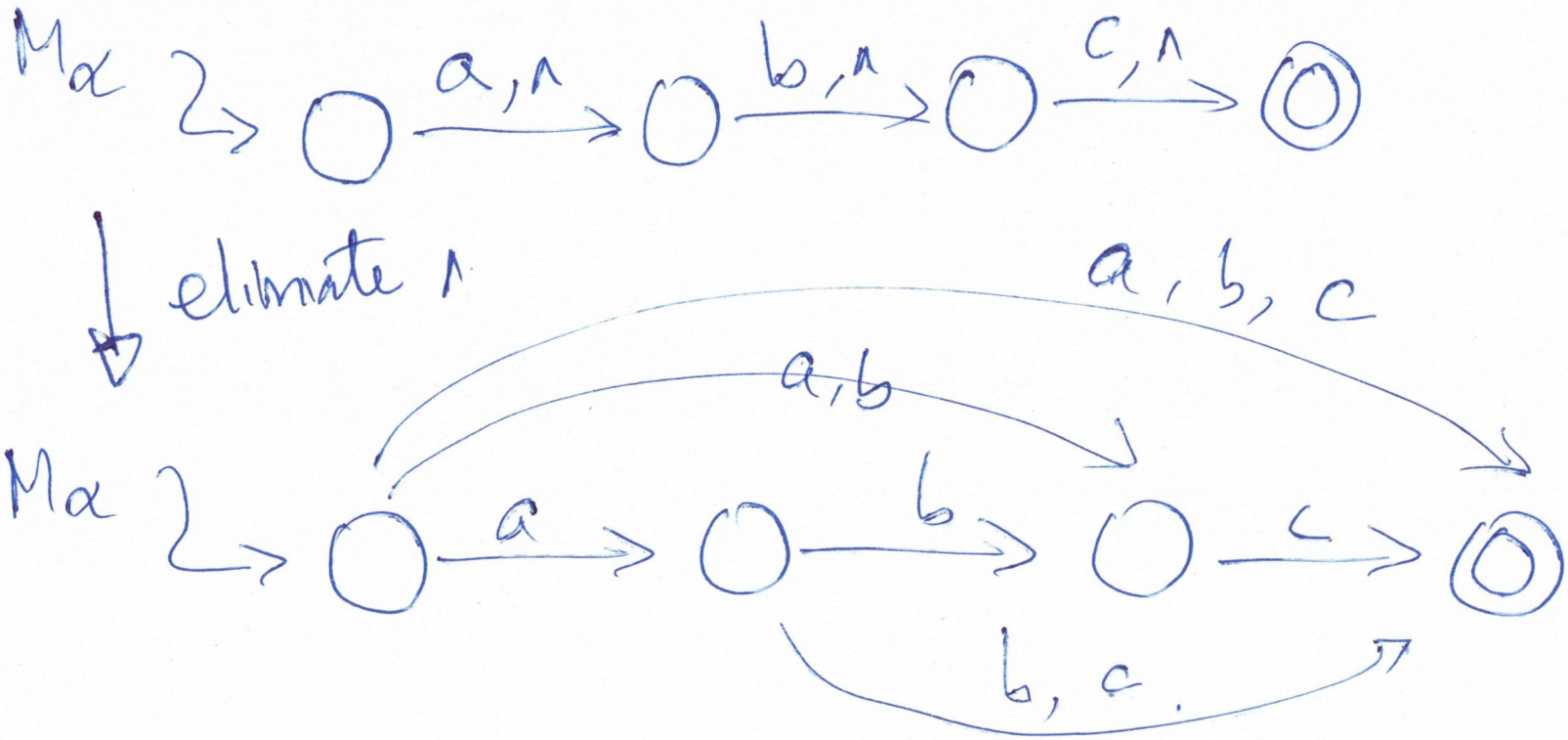
2. Data Structures

FA M_α × FA M_β × FA $M_{pattern}$ = FA M

↕ Cartesian

3. (I have the FA M, already. Recall this M accepts a finite language. Therefore, M has to be a DAG.). Run ~~shortest path~~ longest path alg on the graph of M from init to final and set the word on the path. (Note: the longest path alg is indeed shortest path with -1 weights on edges).

From $\alpha = abc$, how to draw M_α ?



Additional Ex:

Given: a reg lang L_1

a reg lang L_2

a reg Exp (pattern), $(a+b)^*cd$

Want to compute

$$D = \max_{\substack{\alpha \in L_1 \\ \beta \in L_2}} |LCS(\alpha, \beta)| \text{ sat the pattern}$$

In English: to find the max len of longest common subseq between a string in L_1 and a string in L_2 that sat the pattern.

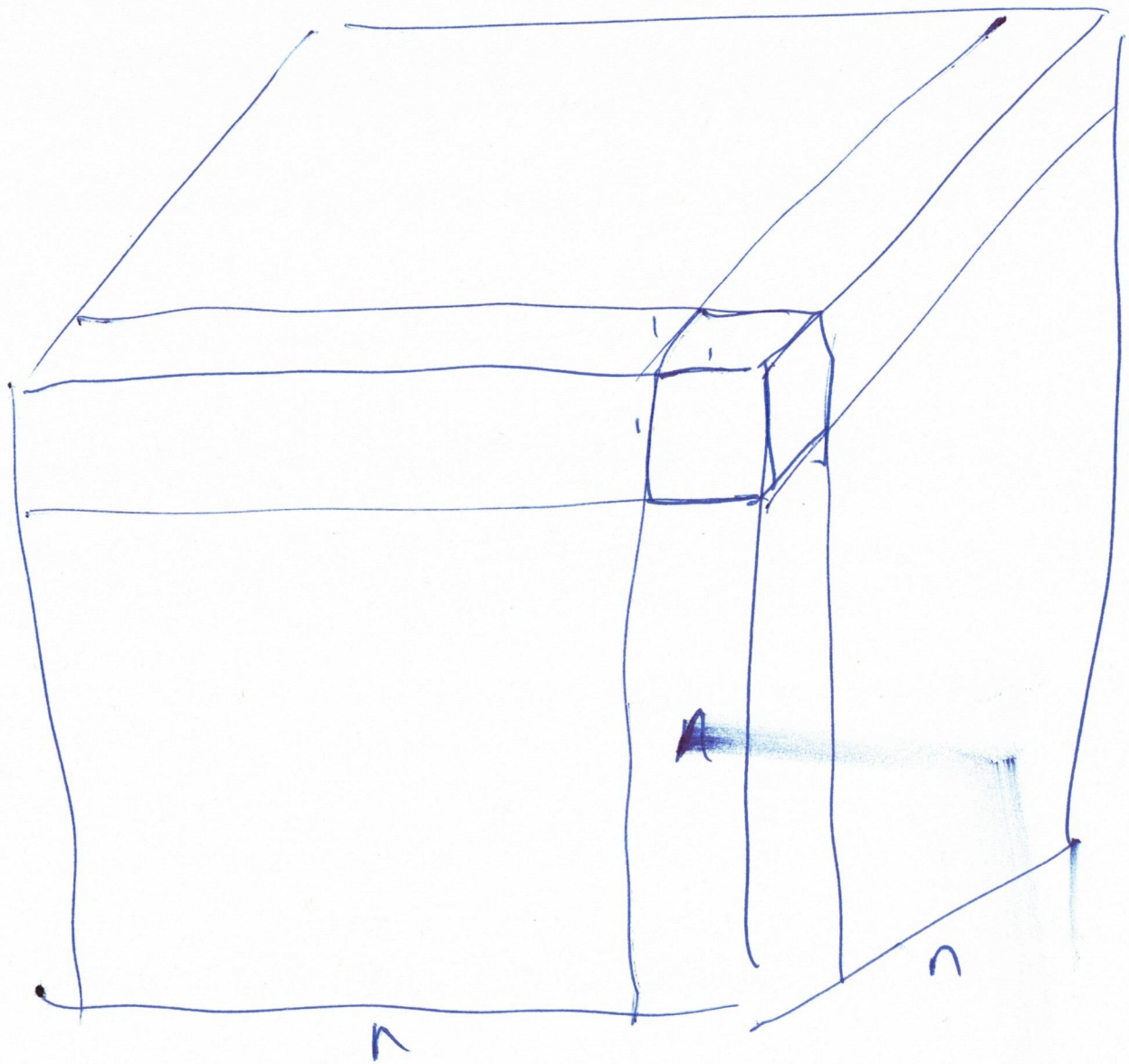
airplane problem. (closest pair with lower bound on dist of two pts.).

~~Step~~ set-up: we have n^3 airplanes, inside $n \times n \times n$ cube. dist between two airplanes ≥ 1.

Goal: linear-time alg for closest pair.

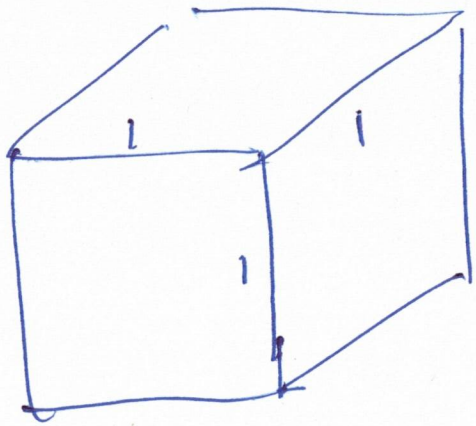
!!!

Remark: Write the reasoning clearly.



I have
n^3 units.

Unit:



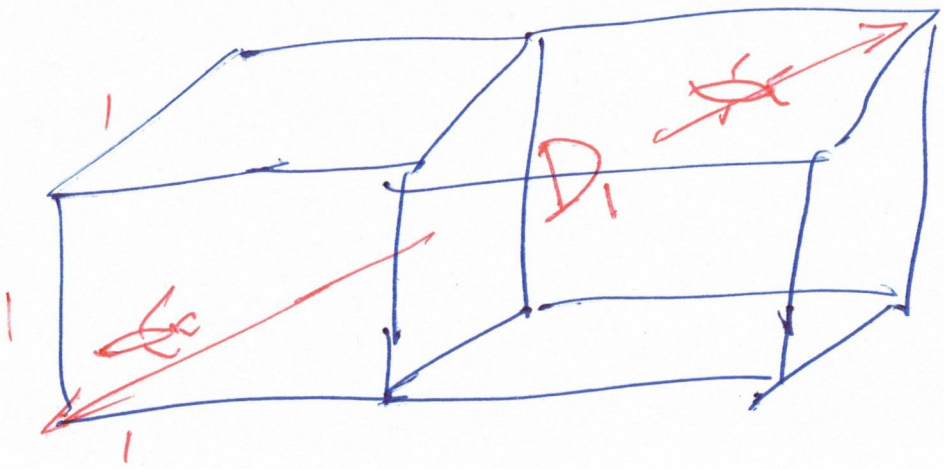
Reasoning: I have n^3 airplanes. I have n^3 units. (Peagon-hole)

Case 1: Each unit holds exactly one airplane,

OR, Case 2: there is a unit holding at least two airplanes.

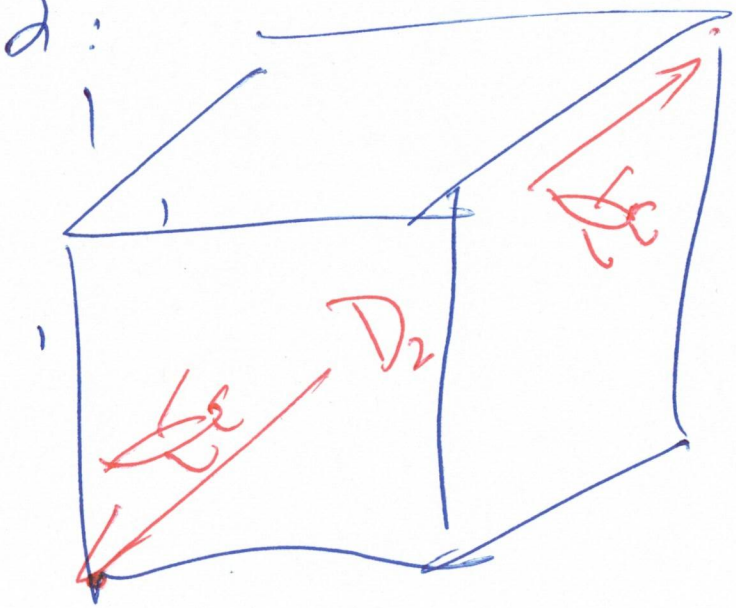
I define D to be the (dist of) closest pair.

For case 1:



$D \leq D_1 = ?$ (middle school math).

For case 2:

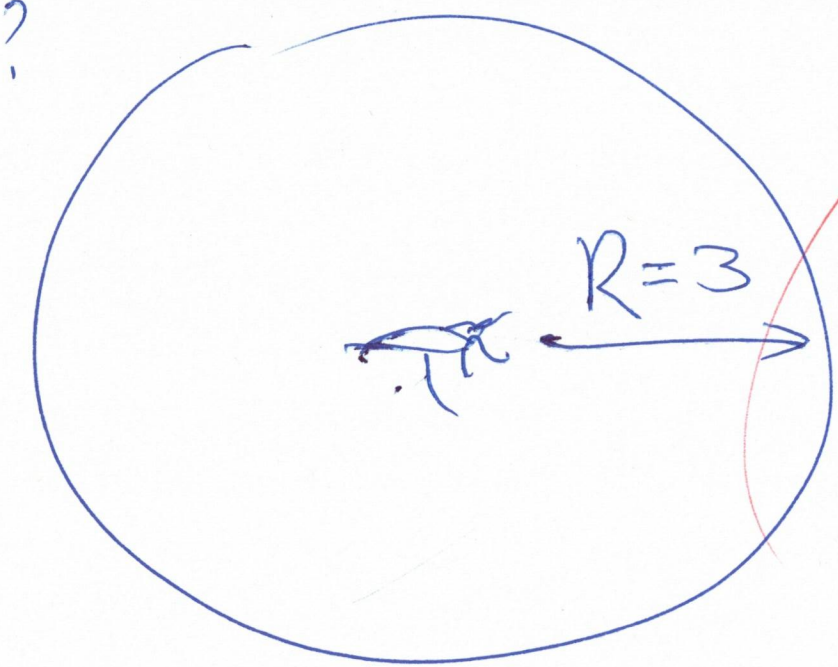


$D \leq D_2 = ?$

In summary, we always have

$$D \leq \max\{D_1, D_2\} \leq 3.$$

For each airplane ~~c~~, I search within radius 3 and see how many airplanes within the range?



You have to figure out: how many airplane could possible fix into this rage?

Key step!

Alg: For each airplane  ,

I put pair ( , )

into a Bag for each

 staying within radius 3.

Get the closest pair in the Bag;

Remark: We use "buckets" trick for implementation.

map an airplane (1.32, 4.7, 2.88)



a unit (1, 4, 2).

I will write another problem (similar to this) in Exam.

BDD Project. (if you code the project by yourself, you should be able to solve this problem in Exam.).

Example. I have a graph with 4 nodes.
R is a BDD (or a Boolformlas) to rep. the graph.

① How many variables?

$R(xx_1, xx_2; yy_1, yy_2)$.

2 vars to rep 4 nodes.

($\because \log_2 4$).

meaning of the R? One-step reachability

$(xx_1, xx_2) \longrightarrow (yy_1, yy_2)$ in the graph

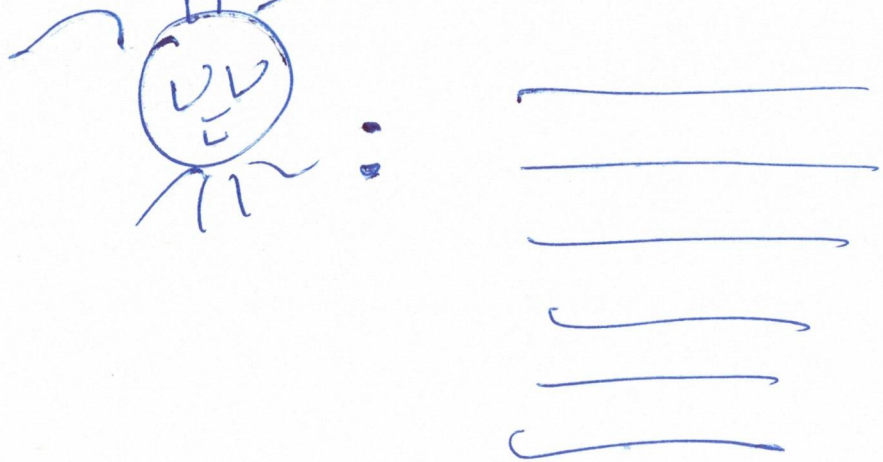
② Code in python for computing R_2 from G (not pseudo-code)

The R: R_2 is to encode all

$\langle v, v' \rangle$ sat. $\exists v''.\ v \to v''$ and $v'' \to v'$.

($v \to v''$: edge in G; $v'' \to v'$: edge in G)

(R_2 is 2-step reachability).



I could also ask you to code R^* (fixed pt, or transitive closure)